

RISHIKA PATEL

+91 7985925729 | rishika.patel2419@gmail.com | linkedin.com/in/rishika-patel | github.com/Rishikaaz

SUMMARY

Performance-driven CS undergraduate specializing in Scalable Backend Development and System Design. Expert in DSA, Python/C++ microservices, and cloud infrastructure (AWS). Proven track record in building secure, containerized applications and solving complex algorithmic problems.

EDUCATION

Vellore Institute of Technology (VIT) B.Tech, Computer Science (CGPA: 8.84)	2023 – 2027
St. Xavier's Convent School Senior Secondary (XII), ICSE	2022
St. Xavier's Senior Secondary School High School (X), CBSE (91.4%)	2020

TECHNICAL SKILLS

Languages: C++, Python, Bash Scripting, C, C#, SQL

Developer Frameworks & Tools: Docker, Docker Compose, Linux, Git, GitHub, ELK Stack

Cloud, Databases & Architecture: DSA, System Design, AWS Services, SQL, REST APIs

Specialized Engineering: Network Protocols (TCP/IP), Microservices, Secure Coding Practices, Threat Modeling

PROJECTS

Containerized HIDS & Distributed SIEM Pipeline

Python, Docker, ELK Stack, Scapy, Linux | May 2026

- Architected a containerized system using Docker Compose to orchestrate ELK microservices, optimizing resource isolation and host overhead.
- Developed a high-throughput Python HIDS sensor using Scapy to capture live traffic and stream real-time JSON alerts over TCP.
- Validated behavioral rules and automated logging by stress-testing pipeline thresholds in distributed network scenarios.

Carbon-Emission-Prediction & Data Pipeline

Python, Scikit-Learn, Pandas, NumPy | Jun 2025 – Jul 2025

- Built an end-to-end predictive pipeline using Python and Scikit-learn to forecast environmental metrics from industrial datasets.
- Implemented feature engineering and data normalization pipelines to achieve highly accurate algorithmic predictions.
- Designed visualization modules to convert multi-dimensional software logs into actionable analytical insights.

AI-Powered Behavioral Anomaly Detection

Python, Scikit-learn, Pandas, Scapy, Networking | ML & Cybersecurity Project | Jul -2026

- Designed an end-to-end threat detection pipeline that captures raw network packets using Scapy, converts them into behavioral features, and passes them to unsupervised models (Isolation Forest, Autoencoders).
- Trained the system on baseline network traffic to identify subtle zero-day anomalies and multi-stage attack patterns that traditional signature-based rules miss.
- Streamed real-time JSON alert logs directly into SIEM workflows, cutting down false positives and enabling immediate incident response.

TECHNICAL EXPERIENCE

Green Skills with AI Technologies Intern

Edunet Foundation | Jun 2025 – Jul 2025 | Remote

Optimized AI-driven solutions for processing large-scale environmental datasets and streamlined data collection workflows.

Cyber Security and Ethical Hacking Intern

Tamizhan Skills | Jun 2025 – Jul 2025 | Remote

Conducted vulnerability assessments on backends, identifying logic flaws and implementing secure coding remediation strategies.

ACHIEVEMENTS

- Solved 300+ LeetCode problems and actively competed in Codeforces contests with a 852 contest rating.
- Selected for the DSCI Cyber For Her Hackathon, qualified for Round 2 of Scripted By{Her} 2.0 Meesho Hackathon and the APCSIP-2026 for a Cyber Cell internship.
- Ranked in the top 2% globally on TryHackMe with a 177-day learning streak in penetration testing and digital forensics.
- Placed 355th out of 5,947 global competitors in the TryHackMe Industrial-Intrusion CTF.
- Secured 7th rank nationally in the Shell n'Zen CTF, solving complex web security and reverse engineering challenges.

CERTIFICATIONS

- Blockchain & Applications (NPTEL, IITK)
- Computer Networking (Coursera)
- Cyber Threat Management (CISCO)
- AWS Cloud Practitioner